Enterprise Routing & VLSM Architecture

Multi-Subnet Dynamic DHCP, Inter-Router Serial WAN Links & Static Routing Implementation

CISCO PACKET TRACER LAB SPECIFICATION

Executive Summary & Topology Plan

VLSM Subnetting

Designed a custom IPv4 VLSM scheme utilizing 192.168.10.0 classful space, optimized across 4 variable subnets (/26, /27, /28, /30) to eliminate IP address wastage.

Dynamic Host Config

Configured Cisco IOS DHCP pools (ENG_POOL, SALES_POOL, SUPPORT_POOL) with excluded default gateway reservations for dynamic IP distribution.

Static WAN Routing

Established a high-reliability WAN serial point-to-point interconnect between Router0 & Router1 with bi-directional static routes.

VLSM Subnetting Architecture

Subnet Allocation

Engineering Dept

Assigned 192.168.10.0 /26 (Subnet Mask: 255.255.255.192) — accommodating up to 62 host addresses.

Sales Dept

Assigned 192.168.10.64 /27 (Subnet Mask: 255.255.255.224) — supporting up to 30 host addresses.

Support Dept

Assigned 192.168.10.96 /28 (Subnet Mask: 255.255.255.240) — supporting up to 14 host addresses.

WAN Point-to-Point

Assigned dedicated 192.168.10.112 /30 (Subnet Mask: 255.255.255.252) — WAN link for router-to-router connection.

Comprehensive IP Addressing Table

Subnet Name	Network ID	Subnet Mask	Gateway IP	Interface / Role	Status
ENG_POOL	192.168.10.0	255.255.255.192 (/26)	192.168.10.1	Router0 Fa0/0	Active
SALES_POOL	192.168.10.64	255.255.255.224 (/27)	192.168.10.65	Router0 Fa1/0	Active
SUPPORT_POOL	192.168.10.96	255.255.255.240 (/28)	192.168.10.97	Router1 Fa0/0	Active
WAN_LINK	192.168.10.112	255.255.255.252 (/30)	192.168.10.113 / 114	Serial2/0 Link	Point-to-Point

Routero DHCP & LAN Configuration

1

Gateway IP Exclusion

Excluded default gateway IPs 192.168.10.1 and 192.168.10.65 from dynamic pool assignment.

2

ENG_POOL Creation

Created DHCP pool ENG_POOL with default-router pointing to FastEthernet0/0 interface.

3

SALES_POOL Creation

Created DHCP pool SALES_POOL serving subnet 192.168.10.64/27.

4

Interface Activation

Activated FastEthernet interfaces using the `no shutdown` command.

Routero WAN & Static Routing

01

Sales LAN Interface

Configured interface FastEthernet1/0 with IP 192.168.10.65 for Sales LAN segment.

02

WAN Interface

Configured WAN interface Serial2/0 with IP 192.168.10.113 255.255.255.252.

03

Static Route Injection

Injected static route pointing to remote Support LAN: `ip route 192.168.10.96 255.255.255.240 192.168.10.114`

04

Serial Link Verification

Verified Serial link protocol changed state to UP.

Router1 DHCP & Interface Setup

1

Hostname & Exclusion

Configured hostname Router1 and excluded gateway address 192.168.10.97.

2

SUPPORT_POOL Definition

Defined dynamic pool SUPPORT_POOL serving network 192.168.10.96/28.

3

Local Gateway & WAN Interface Assignment

Assigned IP 192.168.10.97 to local gateway FastEthernet0/0. Assigned WAN interface Serial2/0 IP 192.168.10.114/30.

Router1 Return Path Routing

Engineering Network Route

Added static route back to Engineering network: `ip route 192.168.10.0 255.255.255.192 192.168.10.113`

Sales Network Route

Added static route back to Sales network: `ip route 192.168.10.64 255.255.255.224 192.168.10.113`

Bi-Directional Symmetry

Guaranteed complete bi-directional routing symmetry across the WAN link.

End-to-End Connectivity Test

Test Execution

Executed end-to-end ping test from endpoint PC10.

Confirmed operational routing table lookup and ICMP path stability.

Ping Results

- Pinged remote gateway 192.168.10.97 across WAN link: **0% loss, Avg = 13ms**
- Pinged local default gateway 192.168.10.65: **0% loss, Avg = 0ms**

Key Network Performance Metrics

100%

ICMP Reachability

0%

Packet Loss Rate

3

Configured DHCP Pools

3

Active Static Routes

Implementation Methodology

01

Subnetting

Calculated VLSM block ranges and subnet masks

02

DHCP Services

Configured IP exclusions and pool definitions

03

Interface Setup

Assigned FastEthernet and Serial IP interfaces

04

Static Routing

Configured next-hop routes on Router0 and Router1

05

Verification

Executed cross-network ICMP ping testing

Conclusion

Lab Complete!

Subnetting

All subnets verified successfully

DHCP

All DHCP pools configured and validated

Routing

Static routes verified successfully

Connectivity

ICMP tests passed across the topology

Topology Status: Fully Operational